

LEARNING OUTCOMES!

Candidates should be able to:

- understand the different types of forest and identify, on a map, their main locations
- understand the physical factors that control the distribution of the different types of forest, and the human factors which have reduced their extent
- understand the reasons for: - the development of plantations in the Indus Plain - afforestation on mountain slopes and plateaux
- understand the value of forests as a resource for development, and the importance of their sustainability
- explain the effects of deforestation, such as soil erosion, silting and flooding
- evaluate possible solutions to the problems caused by deforestation.

FORESTS

- ✓ Forests include everything from grassland and small flowering plants to tall trees.
- ✓ Forests are extensive, continuous areas of land dominated by trees.
- ✓ The desired level of forest is 20 - 30% of the total area of a country.
- ✓ Over the period 2000 - 2007 the country's forests are decreased at a rate of 2.1% per year the ninth highest among the world's nation.
- ✓ There are two types of forests; Productive forests or natural forest and Protective forests or man-made plantation

PRODUCTIVE FORESTS	PROTECTIVE FORESTS
It includes all vegetation grown naturally like alpine, coniferous, etc.	It includes vegetation planted by man like riverain.
The vegetation grows in a haphazard manner.	They are linear plantations (grow in rows).
Tree density is high.	Tree density is moderate.
The canopy is close together.	The canopy is further apart.
They have a variety of different species.	They have a single fast growing specie.
The trees are of different age groups.	The trees are of the same age groups.
The trees are not equally spaced and they vary in a height.	The trees are equally spaced and they have the same height.
They have great commercial value.	They do not have much commercial value
They protect the environment by preventing soil erosion and by acting as carbon sinks it cleans the environment.	They protect the environment by preventing from soil erosion and by acting as carbon sinks it cleans the environment.
It provides wildlife habitat.	It provides wildlife habitat.
The trees lower the temperature of surrounding areas.	The trees lower the temperature of the surrounding areas.
It promotes tourism.	It promotes tourism
More diversity.	Less biodiversity

ALPINE FORESTS

- ✓ Natural forests
- ✓ Found at an altitude of 4000 - 4500 m.
- ✓ They cover the areas of Dir, Chitral, Kohistan (Gilgit - Baltistan).
- ✓ A variety of species are found here which includes Silver Fir, Juniper, Birch.
- ✓ The trees are mainly cushion shaped.
- ✓ These are dwarf forests due to severe climatic conditions like lack of liquid water and very short growing seasons in summers.
- ✓ They grow between rocks to shelter them from cold winds, which would freeze the sap in their thin stems
- ✓ They are evergreen
- ✓ They have stunted growth due to less sunlight which therefore decreases the amount of photosynthesis taking place.
- ✓ They have short growing season as the temperature mainly remains below freezing point.
- ✓ Less biodiversity
- ✓ Grow far apart for competition of nutrients.
- ✓ They are mainly used to obtain firewood for fuel.
- ✓ The roots of the tree grow lateral to have a good grip since the soil with sub soil frozen and the layer of bad rock upward. So, there roots grow sideways to obtain sunlight.



CONIFEROUS FORESTS

- ✓ They are natural forests.
- ✓ They grow at an altitude of 1000 - 4000 m.
- ✓ They are found in the northern areas, KPK, Rawalpindi, Islamabad, Quetta.
- ✓ The species include fir, spruce, chir, oak, didar, kail, willow, walnut.
- ✓ They are evergreen forests and so they do not provide leaves for humus formation and also they store food in their leaves so they are not annually replaced like other deciduous trees.
- ✓ The leaves are needle shaped, small, thick and leathery which helps to reduce the amount of transpiration as transpiration can be quite rapid in warmer summers due to intense continental heating.
- ✓ The trees have the conical shape to the sloping branches of the trees prevent snow accumulation.
- ✓ The thick trunk of a tree is covered by a thick bark is then protected by a resonance material to protect the trunk from cold winds.
- ✓ It is an important source of timber for making furniture and boxes.
- ✓ It protects the environment by conserving soil and by help in checking floods.
- ✓ It is good breeding and conserving center for bird and wildlife, it attracts tourists and promotes tourism industries.
- ✓ It also adds to the scenic beauty.



TROPICAL THORN FOREST

- ✓ They are natural forests.
- ✓ They are found in Punjab plains, Southern and Western Baluchistan, and Sindh Plains.
- ✓ Their main species are acacia, Tamarisk and Salvodra.
- ✓ They have a height of 6 - 10 m.
- ✓ They are forests dominated by thorny hardwood.
- ✓ They have long roots for search of water.
- ✓ The vegetation is scarce due to shortage of water.
- ✓ They are evergreen (to save the energy for drought conditions)
- ✓ It controls soil erosion
- ✓ They are also used as firewood. Also used for animal fodder and grazing grounds for animals.



SUB TROPICAL SCRUB FOREST

- ✓ They are natural forest
- ✓ They are found on the hills and foothills of Himalayas, Suleman, Kirthar ranges, Waziristan Mountains (Peshawar, Waziristan, Kohat, Mardan) and above the Makran Coast range.
- ✓ The main species of this forest are Juniper, Chestnut and Oak.
- ✓ They have long spreading roots.
- ✓ They are round thorny bushes.
- ✓ They are leafless.
- ✓ They are stunted growth.
- ✓ They are sparse due to lack of water.
- ✓ It provides wood for furniture and tools.
- ✓ It supplies firewood. Also used as watershed protection and for grazing purposes.



DESERT VEGETATION

- ✓ Natural forests
- ✓ Found in deserts like Thal, Thar, Nara, Cholistan.
- ✓ Cactus and a variant of other species are found here.
- ✓ They are also known as halophytes as they survive in salty conditions.
- ✓ It has few or no leaves.
- ✓ Sparse (Few Trees and scattered)
- ✓ Cushion Shaped
- ✓ Even height
- ✓ They are important for camel grazing.
- ✓ They also provide firewood.
- ✓ Long well-spaced roots to get moisture from deep underground.
- ✓ Shallow wide spreading roots to catch as soon as it falls before it evaporates.
- ✓ Halophytic because high evaporation leads to saline salts.
- ✓ Seeds dormant and germinate during rain.
- ✓ Fleshy stem to store moisture of cactus.
- ✓ Widely spaced together from large area after rain.



RIVERAIN/BELA FOREST

- ✓ Found on River Indus on active flood plain.
- ✓ Many species are babul and shisham.
- ✓ Linear plantation
- ✓ High yielding commercial hardwood species.
- ✓ Provides wood for making furniture, agriculture implements.
- ✓ Obtain water from the rivers due to high water table.
- ✓ They are evergreen.
- ✓ Stop soil erosion and also protects soil and settlements from flood water.



IRRIGATED PLANTATIONS

- ✓ Man-made trees are planted.
- ✓ Planted in rows and are linear.
- ✓ They are equally spaced.
- ✓ Single species
- ✓ Replanting after cutting
- ✓ Planted near rivers, canals
- ✓ Dependent upon water supplied by man through irrigation.
- ✓ Found in Changa Manga near Lahore, Wan Bachran in Thal Desert, Chichawatni in Sahiwal District and Ghulam Muhammad and Gudu Barrages.
- ✓ Economically important species are planted in large blocks of the same species, Babul, Shisham and Eucalyptus are normally preferred.
- ✓ Important sources of timber, firewood.
- ✓ When planted in linear form provides shade.

Why are there irrigated plantations in the Indus Plain?

- ✓ Construction
- ✓ Firewood
- ✓ Furniture
- ✓ Agricultural implements (Wooden Plough)
- ✓ Irrigation available
- ✓ Prevent erosion of banks
- ✓ Reduces air pollution
- ✓ For shade
- ✓ Reduce timber imports

Explain the advantages and disadvantages of developing more irrigated plantations of trees in lowland areas of Punjab and Sindh.

Advantages

- ✓ Timber for construction, fencing, firewood
- ✓ Food for animals / people
- ✓ Leaf fall for humus / fertility
- ✓ Rainfall

- ✓ Shade / shelter/ reduce temperatures
- ✓ Clean air / more oxygen / absorbs CO_2
- ✓ Scenic beauty increases so promote tourism
- ✓ Protection from soil erosion
- ✓ Employment / work (Wood Cutter, Forest Guides)
- ✓ Habitat for wildlife

Disadvantages

- ✓ High investment needed
- ✓ Long term investment / not a quick profit
- ✓ High maintenance cost
- ✓ Need for irrigation so pressure on water supplies
- ✓ Loss of land for housing / farmland / other land use



MANGROVE FORESTS

- ✓ Natural forests
- ✓ Found along the coastal areas in the Indus and Hub delta near Karachi.
- ✓ They have broad leaves which are waxy and leathery to reduce transpiration.
- ✓ In well water conditions it grows to height of 6 - 8 m otherwise they grow till 3 m.
- ✓ Grow on mudflats
- ✓ Survive in sea water
- ✓ Roots bend into water
- ✓ Roots filter salt from water
- ✓ The species which are common are timar [*Avicennia Officinalis*] Chauri or Kirani [*Ceriops*] and Kinni [*Rhizophorous*], Bgayara.
- ✓ We can only find limited species because of water pollution and due to lack of fresh water availability.
- ✓ Mangroves provide breeding ground for fish and shrimps because the roots of the mangroves trap the fish nutrients brought down the River Indus and they protect the young ones from predators, mud flats are ideal for crabs and shallow water for lobsters.
- ✓ They act as a barrier from huge waves produced by tropical cyclone and tsunami.
- ✓ They also protect the erosion of coastal areas.
- ✓ They are important for migrating birds.
- ✓ It provides wood for the locals for firewood and for construction.
- ✓ It provides fodder for livestock.

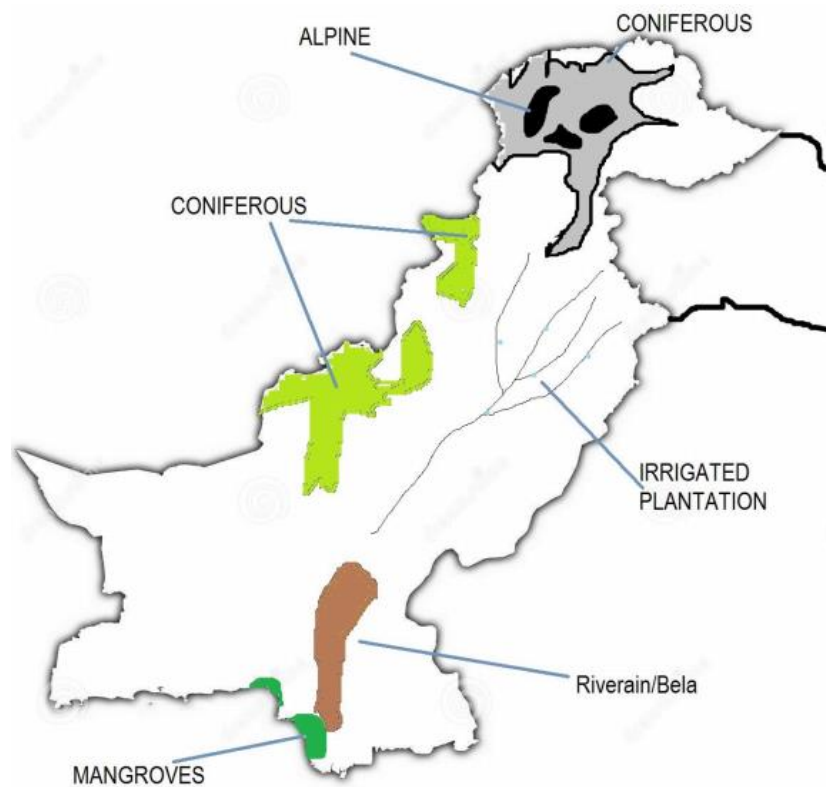
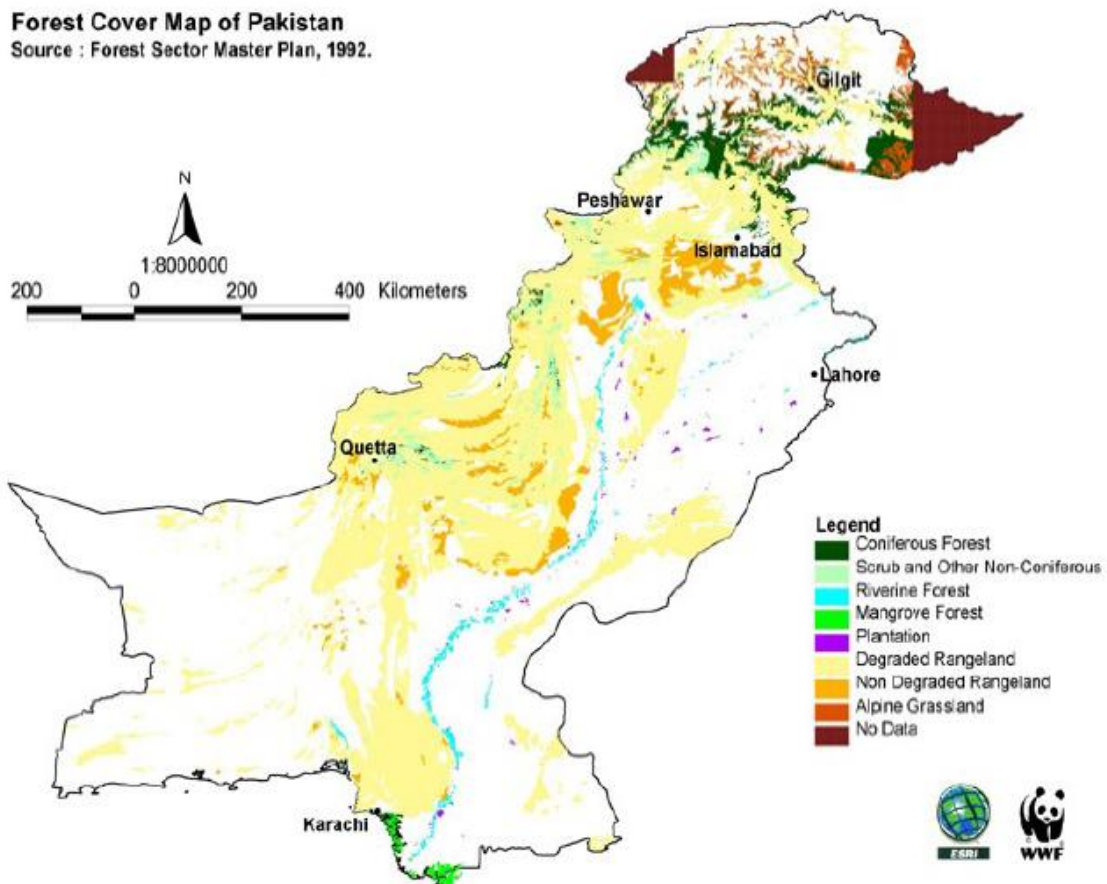
Importance to Fisherman and Fishing villages

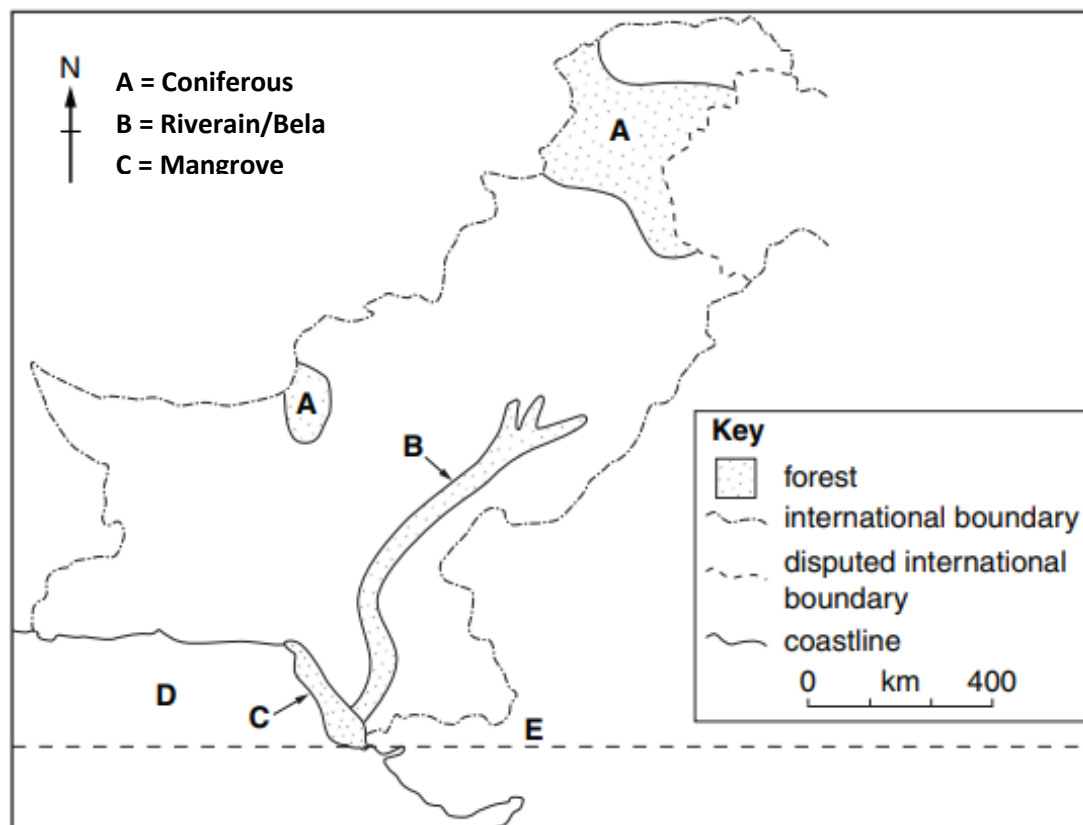
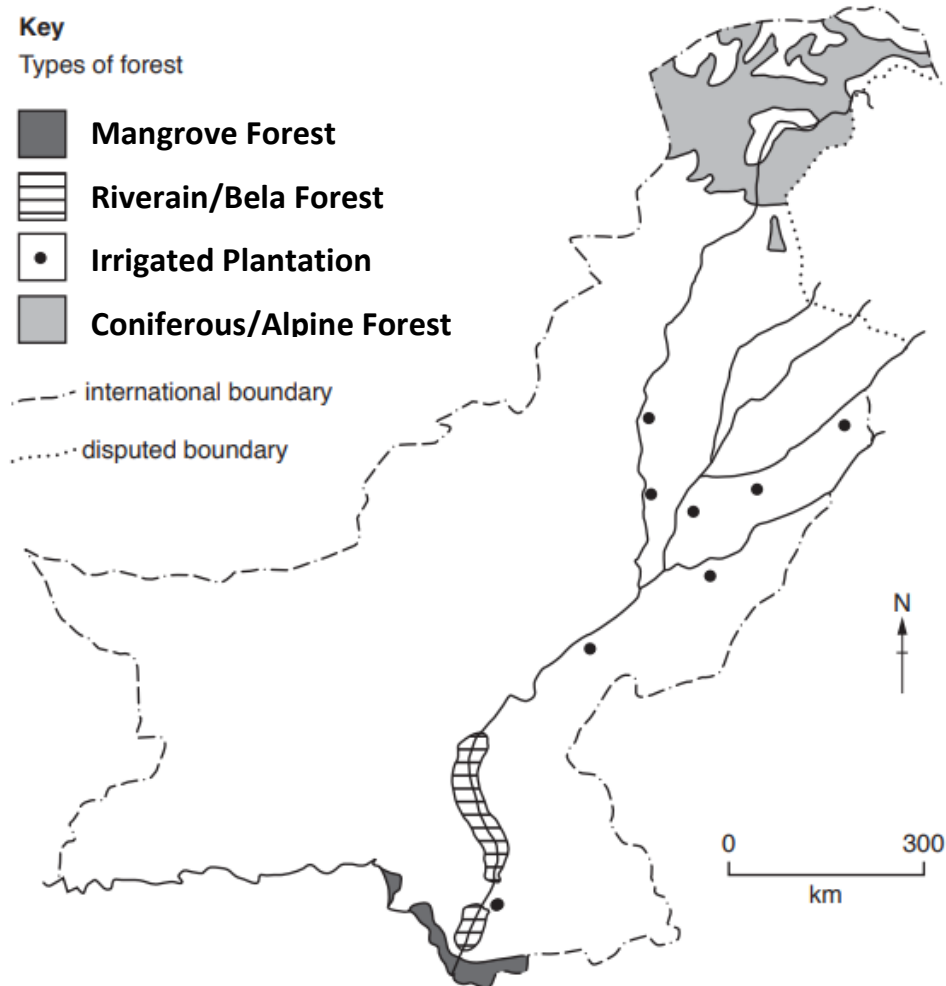
- ✓ breeding area for fishes so many fish there
- ✓ source of income
- ✓ protection to villages (against storms, floods, tidal waves etc.)
- ✓ firewood
- ✓ fodder / food for livestock
- ✓ timber / wood for boats, houses etc.

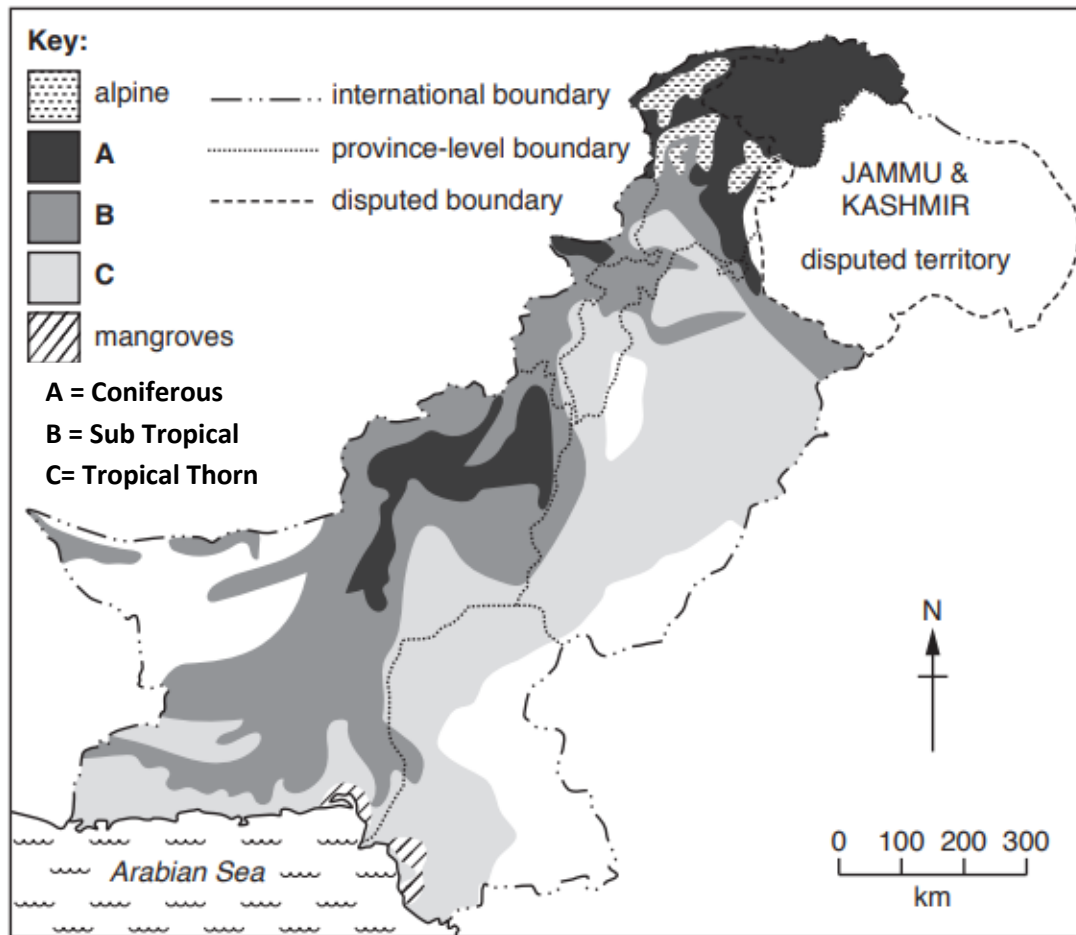


DISTRIBUTION OF FORESTS

Forest Cover Map of Pakistan
Source : Forest Sector Master Plan, 1992.







PHYSICAL FACTORS WHICH INFLUENCE THE DISTRIBUTION OF FORESTS

- ✓ Rainfall;
- ✓ Temperature - mild for non-coniferous / cool for coniferous;
- ✓ Extent of water supply other than rainfall;
- ✓ Soil - soil type influences density of forest;
- ✓ Altitude - types of forest vary with altitude

IMPORTANCE OF FORESTS

- ✓ Natural habitat for wild life.
- ✓ Ensure food supply by protecting soil fertility.
- ✓ Regular water supply to rivers to reduce flooding. It controls floods by binding the soil together and preventing the river beds to be raised.
- ✓ Timber for construction.
- ✓ Plants are used for medical purposes in pharmaceutical industries.
- ✓ Supply of fruit
- ✓ Employment for people in forest based industries.
- ✓ Controls air pollution as the trees acts as carbon sink i.e. they absorb carbon dioxide and gives oxygen.
- ✓ They prevent soil erosion as the roots of trees binds the soil particles together.
- ✓ The stems and the leaves provide shelter from rain and prevent larger splash.
- ✓ Deciduous trees shed their leaves forming humus so adds soil fertility and prevents excessive leaching of nutrients (Leaching is when the nutrients to dissolve in water which goes in the sub surface and therefore less nutrients are available for the roots to absorb).
- ✓ Raw material for wood based industries. Provides wood for making agricultural implements like wooden plough, fences.
- ✓ Roots of the trees help recharge ground water.
- ✓ Trees attract rainfall through transpiration.
- ✓ Wood can be used to obtain pulp which can be further processed into paper.
- ✓ Encourages tourism and helps earn income for the locals.

FOREST AS A SOURCE OF INCOME

- ✓ Fire wood, Charcoal
- ✓ Provides raw material, timber ephedra, fruits nuts
- ✓ Cottage, Small scale industries, pharmaceutical industries
- ✓ Furniture, toys, souvenir sold to tourists
- ✓ Forest attract tourism
- ✓ Creates employment examples: forest guides, rangers.

MINOR FOREST PRODUCTS

- ✓ Resin: is a fluid in the tissue of a plant chir that becomes solid on exposure to the air. It is used for making vanishes, chir pine forest occur in Punjab, Azad Kashmir and KPK.
- ✓ Mazri: used for making mats, baskets and for packing purposes, mazri shrubs is a hardy fiber plant belonging to a group of palms which grow in low, arid, mountainous areas of Suleiman Ranges, Waziristan.
- ✓ Ephedra: is a medicinal shrub used by pharmaceutical factories for treatment of asthma, hay fever and common cold and they are found in Baluchistan.

DOMESTIC USES OF WOOD

- ✓ Firewood for heating
- ✓ Cooking
- ✓ Home building
- ✓ Furniture
- ✓ Fencing

USES OF WOOD IN INDUSTRY AND TRANSPORT

- ✓ Construction of buildings and bridges.
- ✓ Transport railway sleepers, bridges, carts
- ✓ Chemical such as resin, mazri and ephedra
- ✓ Farms such as fences, gates, implements
- ✓ Paper production

IMPORTANCE OF FOREST IN CRAFT INDUSTRY

- ✓ Raw material for wood based industries. Provides wood for making agricultural implements like wooden plough, fences.

IMPORTANCE OF FOREST IN TOURISM

- ✓ Scenic Beauty increases.
- ✓ Encourages tourism and helps earn income for the locals.

IMPORTANCE OF FOREST IN CLIMATE

- ✓ Controls air pollution as the trees acts as carbon sink i.e. they absorb carbon dioxide and gives oxygen.
- ✓ The stems and the leaves provide shelter from rain and prevent larger splash.
- ✓ Trees attract rainfall through transpiration.
- ✓ Help to reduce temperature.

IMPORTANCE OF FOREST IN SOIL

- ✓ They prevent soil erosion as the roots of trees binds the soil particles together.
- ✓ Deciduous trees shed their leaves forming humus so adds soil fertility and prevents excessive leaching of nutrients (Leaching is when the nutrients to dissolve in water which goes in the sub surface and therefore less nutrients are available for the roots to absorb).

SUSTAINABLE FORESTRY

- ✓ Means that we should manage the resources in such a way as to ensure that we will be able to obtain the things that we want from the forests on a regular basis while conserving the natural environment.
- ✓ The use of forests and forests lands in such a way and at a rate that maintain their biodiversity productivity, regeneration capacity and their potential to fulfill now and in the future relevant ecological, economic and social functions, at local, national and global levels and that does not cause damage to other ecosystems.

HOW TO SUSTAIN FORESTS?

- ✓ Planting trees to fill, replace gaps in forests especially in vulnerable areas such as on slopes.
- ✓ Use dead branches for firewood rather than chopping trees down.
- ✓ Educate and train local people into sustainable ways of use.
- ✓ Plant fast growing agricultural trees crop like oil palms, eucalyptus.
- ✓ Maintain a complete forest cover to prevent soil damage.
- ✓ The tree crops can be used to shelter smaller food crops.

- ✓ Wood needed for other purposes such as fuel can be provided by planting patching of fast growing eucalyptus trees.
- ✓ Selective logging of trees of greatest commercial value.
- ✓ Taking out only mature trees and leaving the rest to grow to full size.
- ✓ Keep forest clearances small so that rapid generation is possible.
- ✓ Fines.
- ✓ Punishment.
- ✓ Awareness.

DEFORESTATION

- ✓ Deforestation is the cutting down of trees by human use for any purpose.

CAUSES OF DEFORESTATION

- ✓ Fuel wood: Most of the people living in the rural areas depends on wood for heating and cooking which results in excessive cutting of forests.
- ✓ Road construction: Due to industrialization and urbanization more demand for better transport facilities are required so network is getting larger and denser so forests have been cut down to make roads.
- ✓ Residential purpose because of the increasing population. Growth of large cities has converted forest areas into settlements.
- ✓ Urbanization i.e. for building industries and airports.
- ✓ Forest are cut down to grow crops to meet the food requirement of growing population and to carry out mining activities.
- ✓ For construction of dams and barrages: In order to this, millions of hectares of land were cleared to provide irrigational facilities.
- ✓ To obtain raw material for paper, railway sleepers, industries for furniture's and making boats.
- ✓ Large domestic and foreign market for wood based product. Wood is used in industries little sports goods. Chip board, hard board, veneer plywood and safety matches. Wood is also required for construction industry. Bus bodies, boats and railway coaches use wood as a result forests are cut down to meet the demand of industry.
- ✓ Overgrazing of land by cattle, goats and sheep's has also converted sub-tropical scrub forest and tropical forest areas in desert.
- ✓ Lack of enforcement of forest rules and regulations.
- ✓ Forest Fires.

EFFECTS OF DEFORESTATION

- ✓ No roots to hold the soil together, no interception.
- ✓ Loss of wood for fuel may have to travel much further to find fuel.
- ✓ Shortages of firewood for heating, domestic use.
- ✓ Lack of leaf fall. Lack of decomposition. Lack of humus Loss of fertility, soil lost only rocks left
- ✓ Less infiltration more run off
- ✓ More exposure to sun and wind.
- ✓ Soil become unfit for agriculture if deposited soils is fertile it improves agriculture eg. Piedmont plains and on active and old flood plains.
- ✓ Siltation in reservoirs because soil exposed and washed into rivers.

- ✓ Landslides, avalanches not held by trees.
- ✓ Blocks roads, building destroyed.
- ✓ Loss of scenery, beauty, shade, leads to decrease in tourism so people lose their jobs.
- ✓ Loss of habitat, species lost, extinction, animals move away, less biodiversity.
- ✓ Less rain, less transpiration
- ✓ Less trees to take carbon dioxide so global warming.
- ✓ Loss of timber for furniture, construction.
- ✓ Rivers levels lower
- ✓ Muddy water undrinkable.
- ✓ Air pollution from burning.

Explain how deforestation can cause soil erosion.

- ✓ Soil is exposed / lack of protection from leaves and branches
- ✓ Surface water flow carries soil away
- ✓ Lack of infiltration into ground
- ✓ Less water absorbed by roots
- ✓ No roots to hold soil
- ✓ No leaf fall to add humus / fertility
- ✓ No new plants grow

How can deforestation affect water supplies?

- ✓ Muddy water undrinkable/polluted
- ✓ Reduced evapotranspiration so less rain
- ✓ Silt in reservoirs reduces storage
- ✓ Silt blocks irrigation channels
- ✓ Irregular flow/ comes in bursts
- ✓ Flooding

MEASURES TO CONTROL DEFORESTATION

- ✓ Supplying irrigation to the deforested areas.
- ✓ Commercial species of trees grow rapidly.
- ✓ Reserving land for fuel wood to save valuable species.
- ✓ Creating awareness by the government organization and NGO's about the importance of forests.
- ✓ Supplying natural gas to areas so that people will not depend on wood for fuel.
- ✓ Use selective cutting methods.

- ✓ Restricting the use of bulldozers and heavy destructive machinery in forests.
- ✓ Enforcement of village forestry program.
- ✓ Village and Urban forestry program by planting trees along road sides, park and open spaces.
- ✓ National parks.
- ✓ Awareness
- ✓ Enforcement of forest rules.
- ✓ Educate people about disadvantages of deforestation.

To what extent are these actions possible in Pakistan?

Possible

- ✓ Successful afforestation projects (of badlands in catchment areas)
 - (Tarbela/Mangla Watershed Project)
 - (Rachna Doab Afforestation Project)
 - (Baltistan 1995 onwards afforestation programme by Agha Khan Rural Support Programme - 830000 trees planted)
- ✓ Awareness programmes about value of forests (run by NGOs)

Not Possible

- ✓ High cost
- ✓ Projects difficult to manage in remote/mountainous areas
- ✓ Security issues
- ✓ Growing population/demand for timber/firewood/land (for housing, industry, agriculture, roads)
- ✓ Less Government priorities
- ✓ Very long term projects

SELECTIVE CUTTING

- ✓ Harvesting of selected singles and group of trees.
- ✓ Cutting out trees that are mature or defective or of inferior kinds to encourage the growth of remaining trees in a forest.

TERRACING

- ✓ Steps are cut into steep hill sides and the front of each flat terrace is edged with mud or stone walls known as bunds.
- ✓ The bunds hold back water and soil.

CONTOUR PLOUGHING

- ✓ It is farming practice of ploughing or planting across a slope following its elevation contour lines.

STRIP FARMING

- ✓ It is when two or more crops are grown in the same field sometime one crop may grow under the shelter of a taller crop for the sake of protection.

AFFORESTATION

- ✓ Plantation of trees in a non-forest area, planting tree in a new area.

REFORESTATION

- ✓ Plantation of trees against deforestation in a forest area.

6 MARKS**Question 1****REVIEW BOOK BY MYM**

Around 5 per cent of the land area of Pakistan is forested. To meet sustainable development targets the amount of land covered by forest needs to increase to 25 per cent by 2030. Read the following two views about possible uses of land in Pakistan:

A

Increased forest cover is an important use of land for the future development of Pakistan.

B

There are more important uses of land for the future development of Pakistan than increased forest cover.

Which view do you agree with more? Give reasons to support your answer and refer to examples you have studied. You should consider View A and View B in your answer. [6]

I agree with view A more.

Why?

- ✓ Protect against soil erosion important for the future of agriculture;
- ✓ Protect areas against flooding important for settlements and industry;
- ✓ A valuable resource for industry; Without forests other industries will not succeed;
- ✓ Help to regulate the climate, reduces air pollution and global warming;
- ✓ Mangroves protect against tropical cyclones / without mangroves industry and housing can be lost;
- ✓ Development of tourism industry